

1 Irrigation strategies under contrasting water-storage and rainfall 2 conditions in the North China Plain

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10 Abstract

11 Efficient irrigation allocation is essential for sustaining grain production while limiting water consumption in the
12 North China Plain wheat–maize rotation. How spatial targeting redistributes production losses, and whether
13 regional water-storage information improves rainfall-based irrigation decisions, remain unresolved. Continuous
14 process-based crop simulations compared uniform and targeted allocations under equal irrigation budgets and
15 evaluated annual storage–rainfall strategies against matched rainfall-only controls during 2014–2025. Growth
16 parameters were fitted to multisite observations; water-use parameters were calibrated against Wuqiao treatments in
17 2016–2018 and retrospectively tested in 2019. At a 50% irrigation reduction, targeting retained $0.73 \text{ t ha}^{-1} \text{ yr}^{-1}$
18 more simulated combined dry grain than uniform cuts across 9.87 million rotation hectares, equivalent to 7.22
19 million t yr^{-1} or 4.87% of uniform production. Mean production was lower than under uniform cuts on 37.60% of
20 mapped area, while crop-plus-fallow evapotranspiration increased by 20.76 mm yr^{-1} ($2.05 \text{ km}^3 \text{ yr}^{-1}$). Across nine
21 years with eligible Gravity Recovery and Climate Experiment (GRACE) observations, strategies selected at 95%
22 and 98% grain targets retained 96.27% and 99.63% of conventional production, respectively. The 95% strategy
23 reduced irrigation by 130.72 mm and evapotranspiration by 36.84 mm per rotation. Matched storage–rainfall and
24 rainfall-only strategies selected identical quotas and produced identical outcomes. These conditional results support
25 joint production and consumptive-use constraints and local production safeguards, while showing that additional
26 storage information provides no decision benefit within the tested quota design.

27 **Keywords**

28 winter wheat–summer maize rotation; irrigation management; crop modeling; GRACE; water availability;
29 consumptive water use

30 **1. Introduction**

31 Irrigation management in the North China Plain must reconcile grain production with water-resource constraints
32 across the winter wheat–summer maize rotation (Yang et al., 2022). Wheat grows through a relatively dry season,
33 whereas maize receives more summer rainfall; soil-water reserves connect their responses to irrigation (Yang et al.,
34 2024). Long-term well-watered observations show that rising grain yield has accompanied increasing actual
35 evapotranspiration (ET), despite comparatively stable reference ET (Zhang et al., 2011). Maintaining production
36 with less irrigation therefore requires attention to both the water delivered to fields and the water consumed over the
37 complete rotation.

38 The timing and location of irrigation supply determine its marginal production return. A sixteen-year experiment in
39 2003–2018 compared seven wheat–maize irrigation schedules and found diminishing production gains as irrigation
40 increased (Yang et al., 2022). A four-year Wuqiao experiment also showed that precipitation conditions influenced
41 the preferred wheat schedule and grain production in the subsequent maize crop (Yang et al., 2024). These
42 responses reflect the sequence of crop demand, rainfall and soil-water carry-over across both seasons (Yang et al.,
43 2024; Yang et al., 2022). At regional scale, irrigation requirements vary among locations and hydrological years
44 (Ren et al., 2022; Wu et al., 2019). Crop-model optimization and irrigation-allocation studies provide approaches
45 for using this heterogeneity to guide management (Jiang et al., 2016; Xiao et al., 2024). Under a common water
46 budget, targeting locations with larger marginal grain returns can improve regional production retention, while
47 changing the distribution of production losses.

48 Assessing these allocation consequences requires consistent crop and water accounting. Field delivery, ET, return
49 flows and storage changes describe different quantities, so a reduction in irrigation does not directly establish an
50 equivalent reduction in consumption (Grafton et al., 2018; Perry et al., 2009). Process-based applications of APSIM
51 and RZ-SHAW in the North China Plain connect crop production with water productivity, field water balance and
52 ET (Chen et al., 2010; Fang et al., 2014). However, transpiration formulation remains a major source of variation in
53 simulated wheat water use (Cammarano et al., 2016). Phenology, canopy, biomass, grain and ET observations

54 therefore constrain complementary processes, while documented irrigation treatments assess their combined
55 response to water supply. Continuous simulation retains the soil-water carry-over needed to compare complete
56 rotation histories.

57 Spatial allocation determines where irrigation is supplied, while antecedent hydrological conditions can inform the
58 annual irrigation budget. Recent rainfall describes water input at the crop-system scale, whereas GRACE total water
59 storage integrates several terrestrial stores over a regional footprint (Save et al., 2016; Scanlon et al., 2016). Recent
60 aquifer recovery under water transfers, abstraction management and changing climate illustrates the broader context
61 of regional storage variation (Long et al., 2025; Xu et al., 2026). Storage information may thus characterize
62 conditions beyond local rainfall. Its practical value for irrigation nevertheless depends on whether it changes the
63 selected action or improves the resulting grain–water outcome relative to rainfall-based decisions.

64 Two connected management questions remain unresolved. First, it is unclear how much grain spatial targeting can
65 retain under an identical irrigation budget, and what consumptive-water costs and persistent local production losses
66 accompany that regional benefit. Rotation-wide accounting must retain soil-water carry-over to assess these
67 outcomes together. Second, the additional decision value of regional storage information beyond antecedent rainfall
68 remains uncertain. A storage indicator can distinguish hydrological conditions without changing an irrigation
69 choice. Comparisons using the same production targets, candidate quotas and monitoring availability are needed to
70 isolate its contribution to management outcomes.

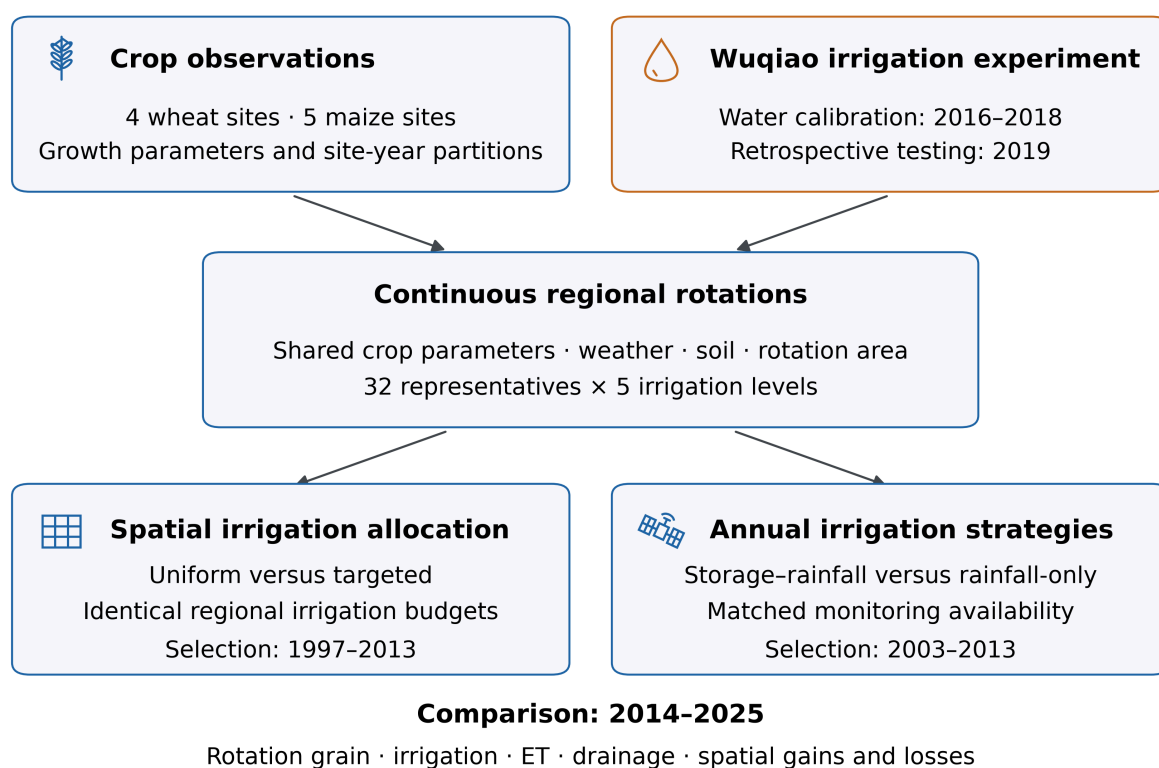
71 The objectives were to (i) quantify rotation grain production, crop-plus-fallow ET and profile drainage under
72 uniform and spatially targeted irrigation allocations with common regional water budgets; (ii) characterize the
73 spatial distribution, magnitude and annual persistence of allocation-related grain gains and losses; and (iii) assess
74 whether antecedent regional storage information changes annual irrigation quotas and grain–water outcomes
75 relative to rainfall-only controls under matched selection and monitoring conditions. Continuous Open Crop Model
76 simulations of the North China Plain wheat–maize rotation, parameterized using multisite crop observations and
77 documented irrigation treatments, provide the basis for these conditional comparisons during 2014–2025.

78 **2. Materials and methods**

79 **2.1. Overall workflow**

80 The framework connects crop-response estimation with spatial allocation and annual irrigation-quota selection
 81 (Figure 1). Multisite crop observations support growth-parameter estimates, while documented Wuqiao treatments
 82 in 2016–2018 constrain eight water coefficients per crop. Wuqiao 2019 provides retrospective testing; station
 83 observations characterize transfer under the recorded management and initialization. Observation coverage and the
 84 distinct estimation and evaluation partitions are summarized in Figure 4. Daily weather, mapped rotation area and
 85 layered soil profiles define 32 regional representatives.

86 The same crop and water parameters drive continuous crop and fallow histories under five irrigation levels. Spatial
 87 allocations selected from 1997–2013 responses share identical regional delivery budgets with uniform reductions.
 88 Annual storage–rainfall and rainfall-only strategies are selected from matched 2003–2013 responses and compared
 89 with common monitoring availability. Grain production, field irrigation, crop-plus-fallow ET and profile drainage
 90 are evaluated during 2014–2025.

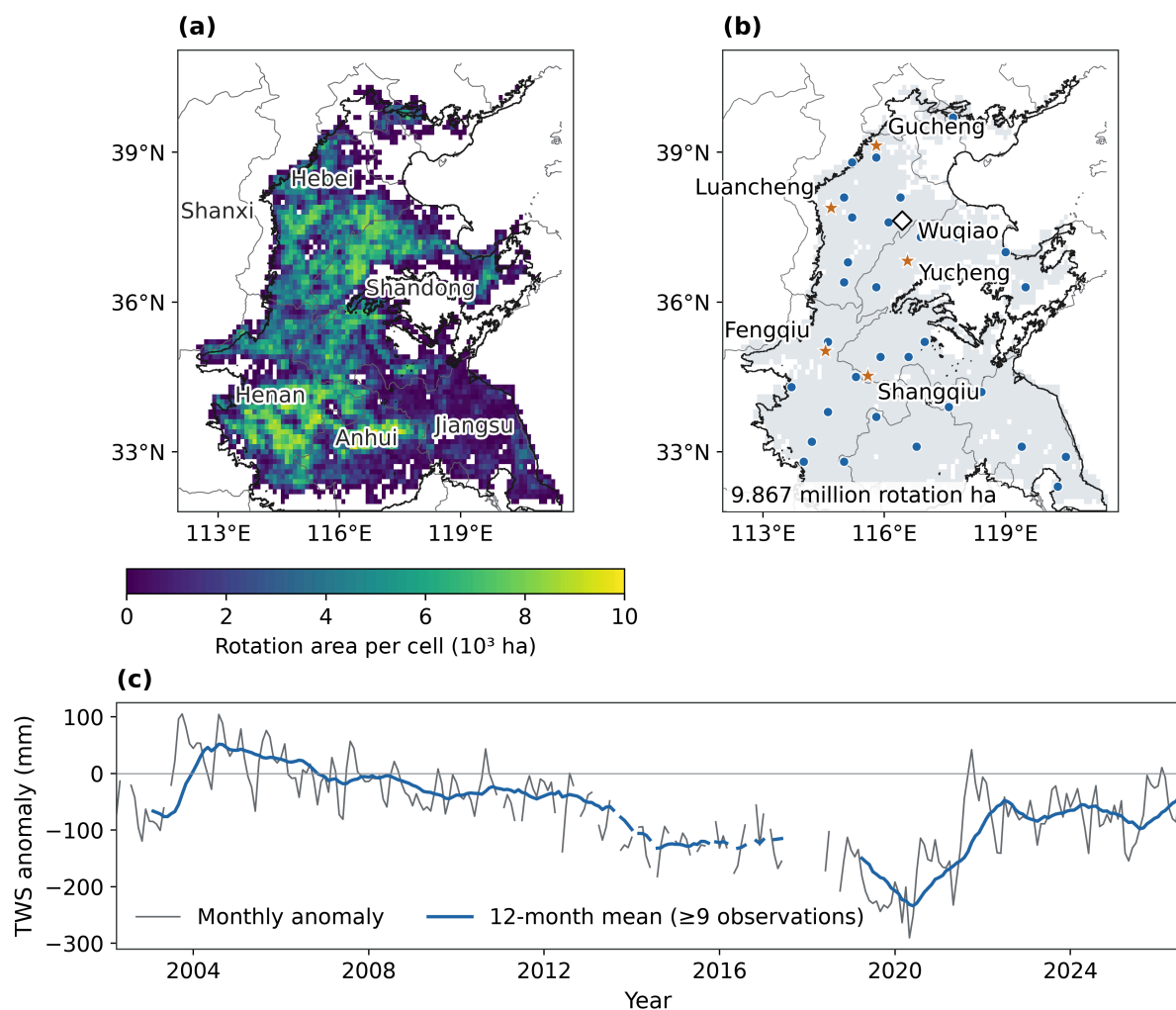


91 **Figure 1. Study workflow.** Multisite observations support growth parameters, while documented Wuqiao treatments calibrate water-
 92 use parameters and provide retrospective testing. Shared parameters drive continuous regional rotations. Spatial allocation
 93 compares uniform and targeted irrigation under identical budgets; annual strategies compare storage–rainfall and rainfall-only
 94 selection under matched monitoring availability. Outcomes cover 2014–2025.

95 2.2. Study area

96 The study area covers the winter-wheat–summer-maize lowlands of the North China Plain within 112–121.5°E and
97 32–41°N, including the Beijing–Tianjin–Hebei sector and the Henan–Anhui–Jiangsu lowlands (Figure 2).
98 Catchment screening, elevation ≤ 200 m and slope $\leq 2^\circ$ define a connected lowland area of 404,658 km². Terrain was
99 represented at 0.01° resolution using resampled SRTM elevation and derived slope (NASA JPL, 2013). The
100 boundary is a reproducible agricultural study-area definition; geographical and terrain-threshold sensitivities are
101 reported in Supplementary Section S3.

102 The 2020 ChinaCP winter-wheat–summer-maize class was corrected by the corresponding CPM cropland
103 percentage to reduce full-pixel area inflation (Qiu et al., 2022). The resulting rotation footprint was 9,866,733 ha
104 before soil exclusions. Three cells with wholly missing soil profiles, totaling 131 ha, were excluded, leaving
105 9,866,602 ha in 3,641 AgERA5 cells. A fixed 2020 footprint applies to all climate years, so modeled totals describe
106 climatic and management responses over a common area rather than historical production trends. The 2015–2021
107 area series, terrain-threshold alternatives and a common-extent comparison with CCD2020 quantify mapping and
108 boundary sensitivity in the supplementary material.



109 **Figure 2. Study domain, rotation area, observation sites and regional water storage. (a) Mapped 2020 wheat–maize rotation area. (b)**
 110 **Five crop-observation sites, the Wuqiao irrigation experiment and 32 regional simulation units. (c) Observed CSR monthly**
 111 **terrestrial-water-storage anomalies and a 12-calendar-month mean requiring at least nine observations. Missing months remain**
 112 **gaps. Thin grey lines show provincial boundaries; black outlines delimit the operational physiographic study domain. Maps use an**
 113 **equiarectangular projection (36°N, 117°E). Provincial boundaries: Natural Earth Admin 1 v5.1.1 (Natural Earth, n.d.).**

114 **2.3. Field observations and environmental data**

115 **2.3.1. Multisite crop observations**

116 Winter-wheat and summer-maize observations were assembled from the National Ecosystem Science Data Center
 117 and linked archives at Fengqiu, Gucheng, Luancheng, Shangqiu and Yucheng. The sources contain long-term crop
 118 monitoring and the Gucheng maize sowing-date experiment. Archived crop-season groups span 2004–2008 at
 119 Fengqiu and Luancheng, 2018–2020 at Gucheng, 2001–2007 at Shangqiu and 1998–2022 at Yucheng. Coverage is
 120 intermittent; exact crop-by-site years and dataset identifiers are listed in Tables S1 and S2.

121 The archive retains 132 crop–plot–season trajectories in 71 original calibration and 61 retrospective-evaluation
122 cases (Figure 4). Quality-controlled targets occur in 127 cases. The 7,397 scored records comprise 795 LAI
123 measurements, 543 in-season aboveground-biomass measurements, 5,675 daily ET measurements, 192 grain-yield
124 samples and 192 harvest-biomass samples (Table 1). Daily ET is available at Fengqiu and Yucheng; grain and
125 harvest-biomass samples are confined to Yucheng. Six quadrats within a crop-year characterize sampling variation
126 rather than six independent growing seasons.

127 Observations and predictions share recorded sampling dates or documented averaging windows. Stage-indexed
128 Gucheng LAI measurements use the corresponding observed stage dates as sampling coordinates; those dates do
129 not prescribe simulated development. Fengqiu ET identifies the lysimeter observation plot. Yucheng ET retains the
130 measurement definitions of its source archive. Provider grain and aboveground dry masses in g m^{-2} are multiplied
131 by 10 to obtain kg ha^{-1} .

132 Eligibility requires valid dates, compatible crop and plot identities, consistent stage sequences and documented
133 quantity units. Wheat LAI above $7 \text{ m}^2 \text{ m}^{-2}$ is excluded under the study-specific screening rule, with raw values and
134 reasons preserved. The six-replicate maize harvest vectors in Yucheng 2012 and 2017 are identical despite different
135 dates and cultivar labels. Both groups remain excluded because their source authenticity is unresolved; 12 samples
136 account for 24 grain and harvest-biomass records. Other observations and all eligible sites remain available.

137 Complete irrigation histories matched to station plots remain unconfirmed. Station growth and ET consequently
138 supply conditional transfer comparisons; no station observation enters the current water-parameter objective.
139 Twenty-six Fengqiu cases retain complete 0–1.5 m moisture profiles from the preceding calendar month: 11 in the
140 original calibration partition and 15 in retrospective evaluation. Volumetric percentages from fifteen 100-mm source
141 layers are conservatively remapped to the model layers without depth extrapolation or clipping. Their monthly
142 timing and subplot support do not establish a measured sowing-day state. Cases without an eligible profile retain
143 their archived initialization.

144 **Table 1. Eligible station-observation records by site and variable.**

Site	LAI	Biomass	Daily ET	Yield	Harvest biomass
Fengqiu	159	163	962	0	0
Gucheng	216	0	0	0	0
Luancheng	123	165	0	0	0
Shangqiu	114	148	0	0	0
Yucheng	183	67	4713	192	192

145 Counts denote individual observations, not independent experiments. Station comparisons retain inherited growth-
146 fitting history and unconfirmed complete management. Yucheng grain and harvest biomass contain six dry-matter
147 samples per crop-year; the duplicated 2012 and 2017 maize groups remain excluded.

148 **2.3.2. Field irrigation experiment**

149 The Wujiao experiment at 37.630°N, 116.444°E contains four wheat irrigation schedules, W0–W3, during the
150 2015–2019 rotation (Yang et al., 2024). Dated 75-mm events, sowing and harvest dates, nitrogen input of 158.4 kg
151 ha⁻¹ per crop and treatment-specific initial total water storage in 2 m are retained. Maize irrigation is identical
152 among treatments within each year: zero in 2016 and 2018, and 75 mm on 18 June in 2017 and 2019. Its W0–W3
153 labels therefore identify preceding wheat management. Seasonal initial water distribution among layers is assumed.
154 Field capacity, wilting point and bulk density are transferred from the published Luancheng profile, with its bottom
155 10 cm extrapolated (Sun et al., 2006). Conductivity follows a separate SoilGrids–Saxton–Rawls prior (ISRIC —
156 World Soil Information, 2020; Saxton and Rawls, 2006); the particle-density prior is 2.75 g cm⁻³.

157 Published seasonal ET equals rainfall plus irrigation plus depletion of 0–2 m storage, with runoff and drainage
158 treated as negligible (Yang et al., 2024). The model represents both outflows. The 2016 maize balance uses 499.1
159 mm of maize-season rainfall; the 599.6-mm rotation total also contains 100.5 mm during wheat. This period
160 reconciliation reproduces the four printed maize ET values within 0.9 mm of rounding, and all 32 crop–treatment–
161 season ET targets remain eligible. Simulations retain their exact archived weather forcing. Wheat crop-start storage
162 follows the pre-sowing simulation, whereas maize is independently initialized from observed treatment-specific
163 total storage. Reported wheat-straw mulch has no measured cover fraction, so the maize simulations retain the
164 exposed-soil evaporation boundary (Yang et al., 2024).

165 Harvest aboveground biomass includes grain and residues cut at ground level and oven-dried to constant mass. Its
166 dry-matter basis requires no moisture adjustment. Published grain yields are standardized to 13% moisture (Yang et
167 al., 2024). Annual 2016–2019 means and standard errors are digitized from the source Supplementary Figure S2
168 and reconciled with its eight printed four-year means. Pixel tolerances are approximately 0.043 t ha^{-1} for wheat and
169 0.056 t ha^{-1} for maize. Observed grain is multiplied by 0.87 for dry-matter fitting; model dry grain is divided by
170 0.87 for comparisons reported at 13% moisture. Annual and four-year means overlap and do not provide separate
171 independent observations.

172 The 2016–2018 harvest years contribute 12 treatment seasons per crop to the water fit, with seasonal ET, dry grain,
173 dry aboveground biomass and annual W3–W0 ET contrasts as targets. The four 2019 treatment means per crop are
174 excluded from fitting and parameter selection. Their prior inspection during model development makes this test
175 retrospective (Figure 4).

176 **2.3.3. Meteorological, soil, crop-rotation and terrain data**

177 Daily AgERA5 v2.0 weather data on the common 0.1° grid supplied minimum and maximum air
178 temperature, precipitation, solar radiation, reference evapotranspiration, vapor pressure and 10-m wind
179 speed for 1996–2025 (Copernicus Climate Change Service, 2020). Reference evapotranspiration (ET_0)
180 entered the water-balance module directly.

181 Native 250 m SoilGrids v2.0 maps of texture, bulk density, organic carbon and coarse fragments were averaged to
182 0.01° , then aggregated over mapped rotation pixels within 0.1° meteorological cells, for six depth intervals between
183 0 and 2 m (ISRIC — World Soil Information, 2020). Hydraulic properties were derived using Saxton–Rawls
184 equations with bulk-density and gravel adjustments (Saxton and Rawls, 2006). Organic matter was approximated
185 using an assumed organic-matter/organic-carbon ratio of 1.724, and air-dry water content was assumed to equal half
186 the wilting-point value. Hydraulic derivation is detailed in Supplementary Section S2.

187 The 500 m ChinaCP map for 2020 identified winter-wheat–summer-maize rotation pixels, which were paired with
188 the corresponding CPM cropland-percentage map (Qiu et al., 2022). Physical classified-cell areas were corrected
189 for projection scale and multiplied by cropland percentages to obtain rotation-area weights. CPM values were
190 sampled at ChinaCP pixel centers after coordinate transformation because the source grids had different origins.
191 Pixel centers assigned these weights to the study domain and 0.1° meteorological cells. The fixed 2020 footprint
192 supplied a common area basis across climate years; the area and mapping sensitivity analyses are described in
193 Section 2.2 and Supplementary Section S3.

194 SRTMGL1 v003 elevation was bilinearly resampled to 0.001° , and slope was derived at that support (NASA JPL,
195 2013). Aligned 10×10 cell means provided elevation and slope on the 0.01° grid used for lowland screening in
196 Section 2.2. Crop-area weights and aggregated soil profiles were linked to daily weather through common 0.1° cell
197 identifiers.

198 **2.3.4. Terrestrial-water-storage observations**

199 CSR RL06.3 monthly GRACE/GRACE-FO total-water-storage anomalies were aggregated using physical cell
200 areas and the official CSR land mask (Save, 2020; Save et al., 2016). The land-mask overlap covers 395,969 km²
201 within the 404,658-km² study area; excluded coastal supports account for 2.15% of its area. The archive contains
202 259 observed monthly solutions from April 2002 to July 2026 relative to the 2004–2009 mean. Source values in
203 centimeters were converted to millimeters, and missing months remain missing. A 12-calendar-month moving mean
204 requires at least nine observations. The 0.25° product grid is finer than the effective GRACE observation scale, so
205 one regional storage indicator is shared across all simulation units (Save, 2020; Save et al., 2016). Soil, surface and
206 other stores were not subtracted; the indicator represents total water storage rather than groundwater alone.

207 The corresponding aggregation without the CSR land mask is retained as a sensitivity analysis. Regional storage
208 changes provide hydrological context for water-management pressures, with no causal attribution to particular
209 irrigation treatments, water transfers or groundwater regulations (Long et al., 2025; Long et al., 2020).

210 **2.4. Crop model description**

211 Open Crop Model (OCM) couples crop growth and a layered soil-water balance at a daily time step (Figure 3).
212 Wheat development combines temperature, vernalization and photoperiod responses; maize development follows
213 thermal time. Development stage (DVS) is anchored at emergence, flowering and maturity. Germination begins
214 when water above the wilting-point storage is available in the represented layer containing the seed; subsequent
215 thermal development starts from that date. Stage coefficients remain fixed, while the soil-water state can alter
216 establishment timing. Recorded or prescribed harvest dates define the management endpoint separately. Soil-water
217 storage passes continuously through crops and fallows.

218 Canopy interception of radiation follows Beer–Lambert attenuation. Aboveground production depends on
219 intercepted PAR, crop-specific RUE, temperature, prescribed nutrition and an empirical power response to realized
220 transpiration supply. PAR is 0.45 of incident shortwave radiation (Kiniry et al., 1989). Stage-specific SLA
221 determines new leaf area, while established cohorts retain their birth area-to-mass ratio (de Wit and Berghuijs, n.d.).
222 Stage-dependent thermal senescence and a separate daily stress-loss fraction determine leaf survival (Tan et al.,

223 2020). LAI emerges from these carbon and area balances without a maximum-LAI parameter. Simulated biomass at
224 wheat flowering or maize silking determines grain number; fixed individual grain mass defines sink capacity, and
225 assimilate supply and wheat stem reserves constrain grain filling.

226 The water balance represents interception, ponding, infiltration, evaporation, root uptake, runoff and bottom
227 drainage. Atmospheric demands are partitioned from reference ET using canopy cover and effective transpiration
228 and evaporation coefficients. Soil evaporation follows a storage-based two-stage reduction (Allen et al., 1998).
229 Root uptake depends on rooted-layer water, local depletion stress and a bounded compensation fraction that
230 redirects unmet demand toward better-supplied rooted layers. Vertical transport uses van Genuchten–Mualem
231 hydraulic functions (Mualem, 1976; van Genuchten, 1980), signed gravity–matric gradients and conservative
232 storage bounds. Relative-arithmetic interface mobility and adaptive internal steps permit wetting and redistribution
233 within 24 daily outer steps. The lower boundary is free drainage with no capillary-rise supply. This explicit layered
234 approximation differs from a full implicit Richards-equation solution. Profile depths follow the supplied inputs,
235 including 2 m for Wuqiao and regional simulations.

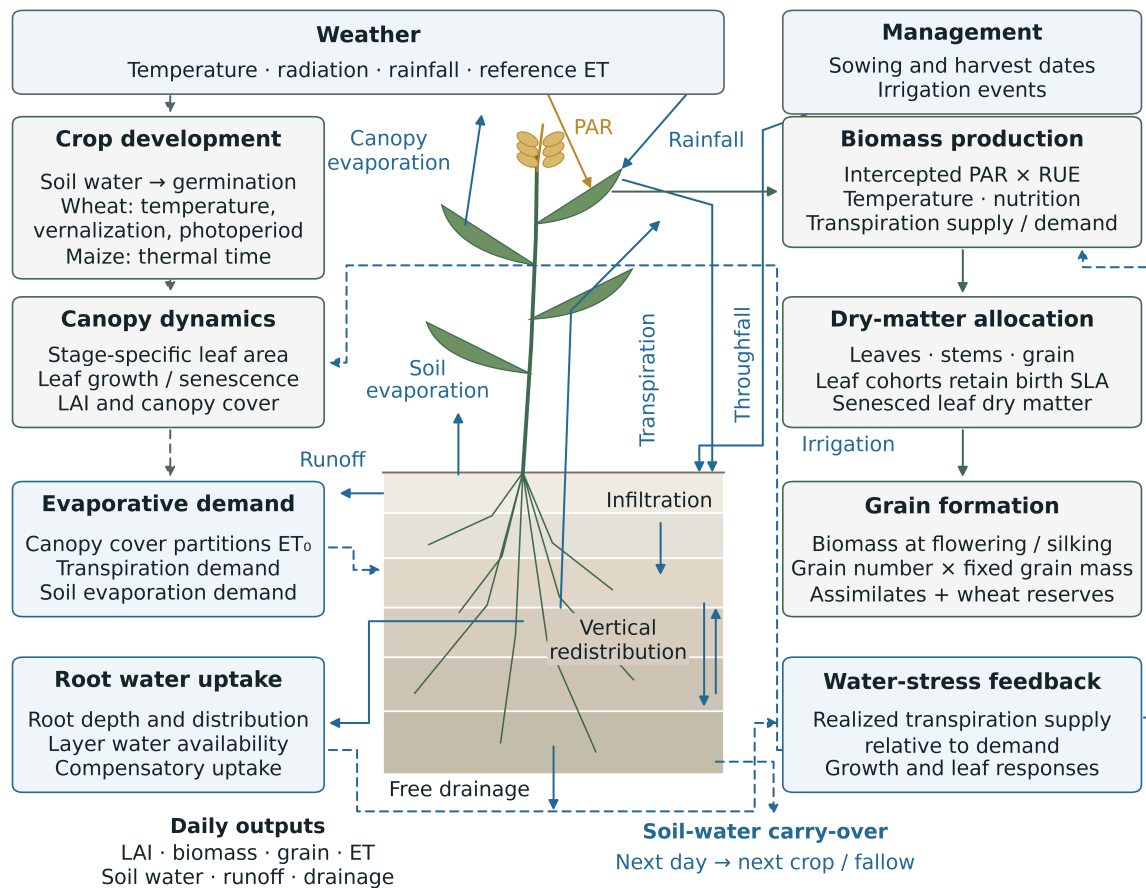


Figure 3. Crop growth and soil-water coupling in Open Crop Model. Gold denotes photosynthetically active radiation (PAR) input; blue solid arrows denote water fluxes; green solid arrows denote dry-matter transfers; dashed arrows denote process controls and feedbacks. Nutrition is prescribed, the lower boundary uses free drainage, and soil water carries through crop/fallow segments. Plant and soil sketches are not to scale.

2.5. Model parameterization, calibration and evaluation

Phenological coefficients, stage-specific specific leaf area and senescence, initial LAI, stem-reserve coefficients and nutrition modifiers follow the crop parameterization described in Section S2. RUE, leaf allocation and grain-number parameters were estimated from 480 wheat and 433 maize growth and harvest observations across four wheat sites and five maize sites. Whole site-years retain their original partitions, with balanced contributions from variables, sites, site-years and crop cases. Station simulations retain their recorded site and cultivar parameterization; Wuqiao and regional simulations use the shared crop parameters (Table S3). Individual grain masses are fixed at 0.045 g for wheat and 0.300 g for maize.

Eight water-use parameters per crop are fitted to the documented Wuqiao treatments in 2016–2018: the transpiration coefficient, soil evaporation coefficient, readily available water fraction, assimilation water-stress exponent, root-density decay, maximum rooting depth, root water-uptake coefficient and root compensation

fraction. Each crop contributes 12 seasonal ET, 12 dry-grain and 12 dry-biomass targets, with three annual W3–W0 ET contrasts derived from the same treatment seasons. RUE, leaf allocation and grain-number parameters remain fixed. Bounded least squares combines scaled observation residuals and weak parameter priors; objective weights, bounds and convergence settings are specified in Section S2.4.

The fitted water parameters are fixed before comparison with the four 2019 treatment means per crop. Previous inspection of these outcomes makes the test retrospective. Station observations have zero weight in this water fit and assess transfer under unconfirmed complete irrigation histories. Regional strategy selection uses historical weather responses; the crop-calibration years overlap the 2014–2025 regional comparison period, so those outcomes are conditional scenarios rather than time-independent crop validation. Observed and simulated quantities retain matched sampling windows and grain-moisture bases.

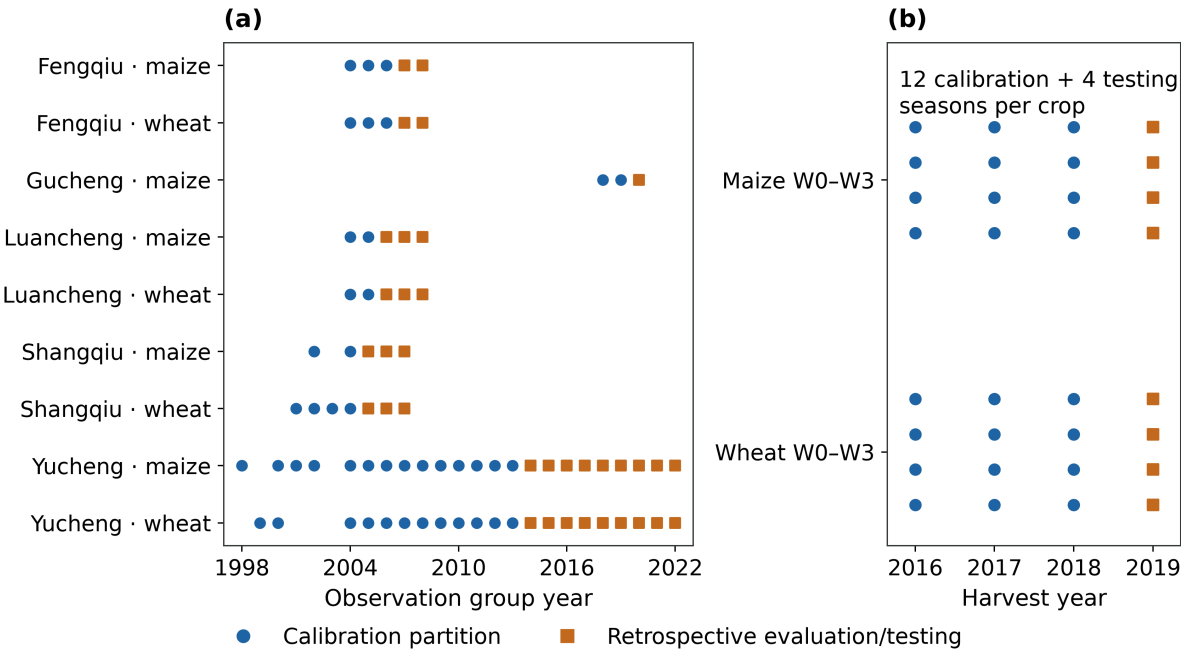


Figure 4. Evidence partitions. (a) Whole station site-years retain their original calibration and retrospective-evaluation partitions; dates denote observation-group years. All five eligible sites are included. (b) Four Wuqiao treatment means per crop-year provide 12 calibration seasons in 2016–2018 and four retrospective testing seasons in 2019. Station comparisons retain earlier growth-fitting history and unconfirmed complete irrigation management.

2.6. Regional upscaling and irrigation scenarios

Thirty-two area-weighted climate–soil representatives were selected from actual source cells using deterministic clustering of latitude, longitude, plant-available water, and 1997–2013 mean precipitation and reference ET. Features were standardized with rotation-area weights. Each cluster uses the actual cell nearest its centroid; the full

269 source-cell mapping and represented areas are retained. Climate years used for policy evaluation do not enter the
270 climate features used for selection. Shared crop parameter sets are transferred regionally without extrapolating site-
271 specific adjustments.

272 Each representative has five continuous histories at irrigation fractions 0, 0.25, 0.50, 0.75 and 1.00. The
273 conventional wheat schedule is 60 mm at sowing on 12 October, 90 mm on 25 March, 90 mm on 15 April and 60
274 mm on 10 May, totaling 300 mm. Maize receives 80 mm at sowing on 15 June. These common schedules are
275 declared scenarios and are not observed universal practice across the plain. Every event amount is multiplied by the
276 irrigation fraction, with its timing fixed. Wheat is simulated through 10 June and maize through 5 October;
277 intervening fallows carry the soil state forward. Initial plant-available water is 80% at the start of the 1996 spin-up,
278 with no subsequent seasonal storage reset. All crop and fallow ET, drainage and runoff are included. The final 2025
279 cycle ends on 5 October because the next fallow lies outside the declared endpoint. Nutrition is represented by
280 prescribed fertilized-crop modifiers; nitrogen cycling and spatially varying nutrient limitations are not dynamically
281 simulated.

282 Eight purposively selected non-representative cells provide paired spatial aggregation checks under all five
283 irrigation fractions. Two cells from each latitude quartile have the largest area-times-feature-distance score. Actual-
284 cell weather and profiles use the same continuous model and crop parameters. Their differences from
285 representative-cell predictions measure crop-model aggregation error rather than field-observation accuracy
286 (Supplementary Section S3).

287 **2.7. Irrigation strategies and comparisons**

288 **2.7.1. Spatial allocation under common irrigation budgets**

289 Uniform allocation assigns the same irrigation fraction to every represented hectare. Targeted allocation
290 chooses area shares x_{jk} for representative j and simulated irrigation fraction k by linear programming. The
291 objective maximizes $\sum_j A_j \sum_k x_{jk} \bar{Y}_{jk}$, where $\bar{Y}_{jk}$ is mean annual wheat-plus-maize grain in 1997–2013.
292 Constraints require $\sum_k x_{jk} = 1$, $0 \leq x_{jk} \leq 1$ and $\sum_j A_j \sum_k x_{jk} I_k = (1 - r) \sum_j A_j I_1$, with reductions
293 $r = 0.25, 0.50$ or 0.75 . Within-representative mixtures describe shares of area receiving actual simulated
294 irrigation treatments; crop responses are not interpolated between irrigation fractions. Allocations remain
295 fixed in 2014–2025. Both policies deliver exactly the same total field-irrigation water in every year, so
296 their differences reflect allocation under the same volume constraint.

297 Spatial contrasts at the common 50% irrigation cut use frozen shares of complete continuous irrigation histories.
298 Their area-weighted grain, ET and drainage differences map from 32 representatives to 3,641 source cells. Fixed
299 southern (32–36°N), central (36–38°N) and northern (38–41°N) bands summarize the distribution. Counts of
300 positive annual contrasts describe consistency within the modeled record (Figure 8; Table S9).

301 **2.7.2. Annual quotas based on antecedent water availability**

302 Relative water-availability classes combine two antecedent indicators: regional total-water-storage status and local
303 rainfall before wheat sowing. The storage indicator is the mean of available August and September GRACE
304 observations in the calendar year preceding harvest, with at least one monthly solution required. Precipitation is
305 accumulated from 1 July to 30 September at each simulation unit, covering 92 days before the prescribed 12
306 October wheat sowing. The corresponding precipitation/reference-ET ratio is retained as a diagnostic of climatic
307 water availability. These antecedent indicators do not use rainfall or crop outcomes from the subsequent growing
308 season.

309 Thresholds use 2003–2013 harvest years, the common selection period with antecedent GRACE coverage. The
310 regional storage median is 13.38 mm, and the rotation-area-weighted precipitation median is 431.74 mm. Values
311 below their respective thresholds define lower storage and lower rainfall; values at or above define higher storage
312 and higher rainfall. Crossing the two indicators gives four hydroclimatic classes. The classes describe relative
313 antecedent water availability, rather than absolute water scarcity derived from water demand, allocation rights or
314 access. Spatial differences within a year arise from precipitation; GRACE does not supply independently resolved
315 field-scale storage classes.

316 For each class, five continuous irrigation histories provide paired grain and ET responses relative to conventional
317 irrigation, weighted by rotation area and year. The primary candidate is the lowest fraction retaining at least 95% of
318 class-mean conventional grain in 2003–2013; a 98% target assesses a stricter production criterion. Selected
319 fractions and class thresholds remain frozen in 2014–2025. Higher-storage classes have no evaluation exposure.
320 Required GRACE observations are missing for 2014, 2018 and 2019.

321 Two annual strategies apply the irrigation fractions selected from fixed-fraction responses at grain-retention targets
322 of 95% and 98%. At each simulation unit, the antecedent storage–rainfall class determines the fraction before wheat
323 sowing. This fraction scales all four wheat applications and the subsequent maize application in the same rotation

year. Application dates, crop parameters, weather, hydraulic profiles and the free-drainage lower boundary are identical to those in the fixed-fraction experiment. Conventional irrigation applies before 2003 and whenever an eligible storage class is unavailable, including evaluation harvest years 2014, 2018 and 2019.

Each strategy is simulated continuously from the 1996 spin-up through 2025. Soil-water states pass between crops and fallows and across annual changes in irrigation fraction. Conventional irrigation supplies the paired continuous reference; replays at two simulation units verify the scenario driver against conventional outputs. Class thresholds and irrigation fractions remain fixed after selection. Outcomes cover all twelve comparison years and, separately, the nine years with eligible classes. Production retention is the ratio of period-mean regional grain production to its corresponding conventional mean; annual ratios describe variation within the period. The strategies assume access to antecedent observation periods, while release availability of GRACE before sowing remains unverified.

2.7.3. Rainfall-only controls and matched monitoring coverage

Rainfall-only controls use the same 2003–2013 simulation-unit years with eligible GRACE observations, the same rotation-area weights, irrigation candidates and grain-retention targets. Selection groups responses by antecedent rainfall alone and chooses the lowest fraction meeting the relevant target. The rainfall threshold is identical to that in the joint classification. Matched controls receive conventional irrigation whenever the storage–rainfall strategy lacks an eligible class, preserving both the selection cohort and monitoring coverage. Continuous differences between these paired strategies consequently assess the incremental effect of storage-based grouping.

Separate rainfall-available controls apply the selected rainfall-only fractions whenever an eligible rainfall group exists, including years without GRACE coverage. Their complete histories assess the combined effect of rainfall-based management and greater monitoring availability. Irrigation changes in earlier gap years can alter the soil-water state of later crops, so these controls are reported separately from the matched comparison.

2.8. Evaluation metrics, spatial statistics and water accounting

Performance is summarized by RMSE, mean bias error, nRMSE relative to observed mean, Nash–Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970) and squared Pearson correlation (R^2). Station scores give equal weight to available sites, site-years within sites, crop cases within site-years and records within cases. Wuqiao treatment means receive equal weights within each crop and partition. Field calibration and retrospective testing remain separate. Grain and harvest biomass in station comparisons are dry matter; Wuqiao grain results retain the published 13% moisture basis, with a single 0.87 conversion between dry and reported grain. Biomass remains dry matter throughout.

353 Seasonal curves group matching observations and predictions into 15-day wheat and 7-day maize bins at
354 observation-window midpoints in days after sowing. Case means receive equal weight within site-years, site-year
355 means within sites and available sites within bins. Standard errors describe variation among contributing site-year
356 means rather than model uncertainty. Site-specific curves use equal site-year weights. Coverage varies among bins,
357 and gaps remain uninterpolated. Accuracy scores use unbinned records. Yucheng harvest error bars describe quadrat
358 sampling variation within crop-years.

359 ET sums use identical observed and simulated measurement days within each case and retain the station balancing
360 weights. They exclude unmeasured days and do not represent complete-season observed totals. Regional grain
361 production uses dry mass and rotation-area weights; actual ET includes crops and fallows. Paired policy differences
362 cover 2014–2025. Daily water and carbon residuals and continuity of carried soil-water storage assess numerical
363 conservation. Field irrigation, ET, drainage below 2 m and groundwater storage have distinct accounting
364 boundaries.

365 Spatial distribution statistics describe targeted-minus-uniform policy contrasts in mean annual combined wheat–
366 maize dry grain and water fluxes over 2014–2025. The 3,641 mapped 0.1° source cells carry outcomes from 32
367 representative response units. Cell weights are the static 2020 mapped rotation areas after cropland-fraction
368 correction; they represent mapped rotation area rather than observed harvested area. Gains and losses exceed $+10^{-6}$
369 and fall below -10^{-6} kg ha⁻¹, respectively; contrasts within these bounds are classified as unchanged. Group means
370 use rotation-area weights, regional contributions sum area multiplied by the corresponding contrast, and weighted
371 quantiles use the inverse empirical cumulative area distribution. Annual loss frequencies use the same sign
372 threshold across the 12 years. Connected loss footprints comprise occupied cells sharing a horizontal or vertical grid
373 edge (rook adjacency). The statistics describe the conditional modeled allocation footprint; mapped cells do not
374 provide independent replication or cell-specific validation.

375 **3. Results**

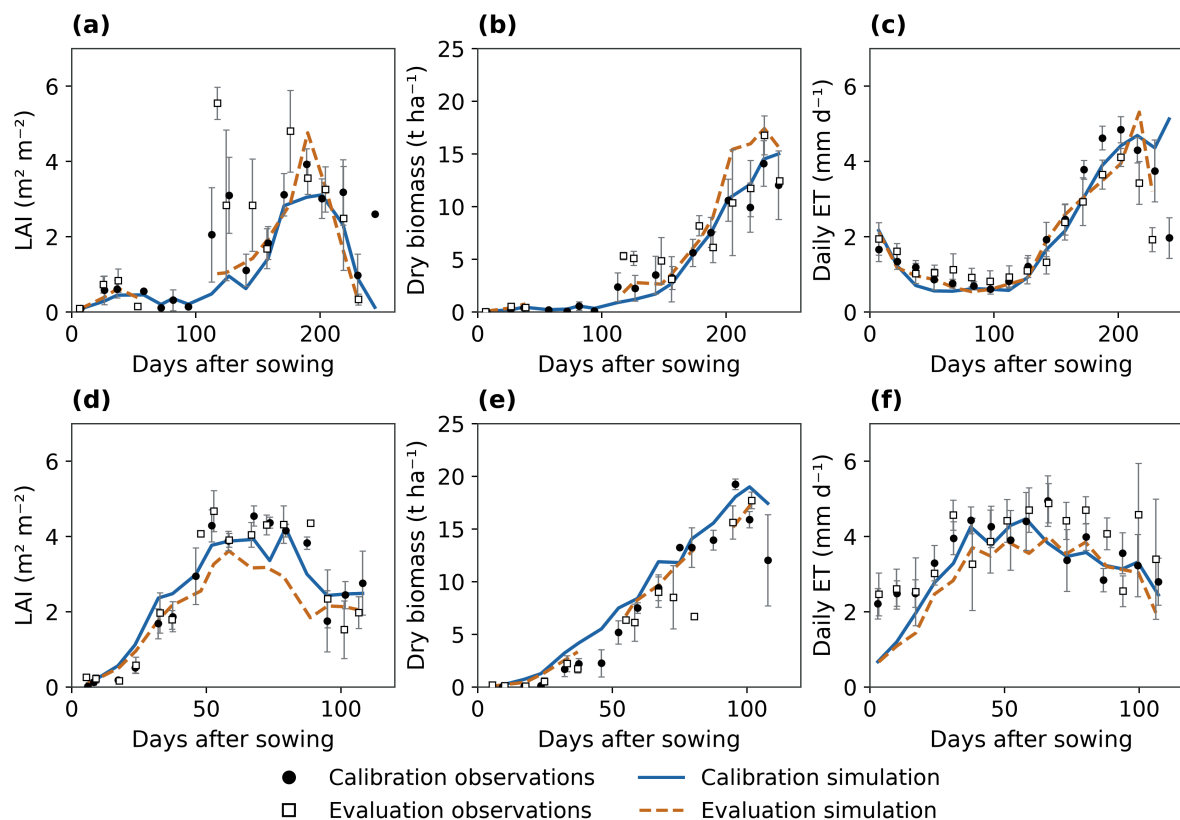
376 **3.1. Conditional multisite crop comparisons**

377 Retrospective-evaluation LAI RMSE was $1.843 \text{ m}^2 \text{ m}^{-2}$ for wheat and $0.995 \text{ m}^2 \text{ m}^{-2}$ for maize, with NSE of 0.184
378 and 0.689, respectively (Figure 5). LAI RMSE in the original calibration partition was 1.727 and $1.138 \text{ m}^2 \text{ m}^{-2}$. In-
379 season biomass RMSE in retrospective evaluation was 4.413 and 2.789 t ha^{-1} , with biases of +1.314 and +1.158 t

380 ha^{-1} and NSE of 0.349 and 0.796. Site-specific seasonal comparisons retain their available observations and
 381 original partitions (Figures S6–S11).

382 Dry-grain RMSE was 1.256 t ha^{-1} for wheat and 1.402 t ha^{-1} for maize in the original calibration partition,
 383 compared with 0.834 and 2.286 t ha^{-1} in retrospective evaluation (Figure S5). Evaluation grain biases were $+0.485$
 384 and -1.586 t ha^{-1} , and NSE was 0.099 and -1.566 . Evaluation harvest-biomass RMSE was 2.736 t ha^{-1} for wheat
 385 and 3.719 t ha^{-1} for maize; the corresponding observation counts were 54 in nine wheat years and 48 in eight maize
 386 years.

387 Daily ET RMSE in retrospective evaluation was 1.809 mm d^{-1} for wheat and 2.572 mm d^{-1} for maize, with biases
 388 of $+0.141$ and -0.734 mm d^{-1} and NSE of 0.121 and 0.012 (Figure 5). ET summed over matched measurement days
 389 had evaluation RMSE of 51.29 mm for wheat and 74.17 mm for maize, biases of $+25.94$ and -55.58 mm , and NSE
 390 of 0.826 and 0.118 (Table 2). These sums exclude days without ET measurements.



391 **Figure 5. Multisite seasonal crop comparisons.** Wheat and maize occupy the top and bottom rows; columns show LAI, aboveground
 392 dry biomass and daily actual ET. Filled and open observations distinguish the original calibration and retrospective-evaluation
 393 partitions; solid blue and dashed orange lines show the corresponding simulations. Error bars describe standard errors among site-
 394 year means. Balanced means use 15-day wheat and seven-day maize bins, and unobserved bins remain gaps. Accuracy scores use
 395 unbinned records. Complete station irrigation management remains unconfirmed.

396 **Table 2. Conditional station ET sums on matched measurement days.**

Crop	Dataset	Seasons	NSE	RMSE (mm)	MBE (mm)
Maize	Prior calibration partition	11	0.511	45.60	-34.92
Maize	Retrospective evaluation	10	0.118	74.17	-55.58
Wheat	Prior calibration partition	11	-0.300	84.94	-23.53
Wheat	Retrospective evaluation	11	0.826	51.29	25.94

397 Observed and simulated sums use identical measurement days within cases, with balanced site, site-year and case
398 weights. Unmeasured days are excluded. Original calibration partitions retain an earlier growth-fitting history;
399 evaluation is retrospective. Complete irrigation-log matching remains unconfirmed.

400 **3.2. Wuqiao calibration and retrospective testing**

401 Seasonal ET calibration RMSE was 40.19 mm for wheat and 37.44 mm for maize across 12 treatment seasons per
402 crop, with biases of -12.65 and +3.60 mm and nRMSE of 8.97% and 12.69% (Table 3; Figure 6). Across the four
403 2019 testing means per crop, ET RMSE was 73.80 and 55.64 mm, bias was +48.93 and -35.90 mm, nRMSE was
404 21.98% and 14.13%, and NSE was 0.503 and -0.346, respectively.

405 At 13% grain moisture, calibration RMSE was 1.249 t ha⁻¹ for wheat and 2.267 t ha⁻¹ for maize; retrospective-
406 testing RMSE was 3.161 and 1.700 t ha⁻¹. Testing grain biases were +3.108 and +1.403 t ha⁻¹, with nRMSE of
407 83.27% and 15.97% and NSE of -4.654 and -3.641 (Table 3; Figure 6). Dry aboveground-biomass RMSE was
408 4.222 and 4.688 t ha⁻¹ in calibration and 5.427 and 2.038 t ha⁻¹ in testing, respectively. Testing biomass nRMSE
409 was 55.66% for wheat and 10.44% for maize.

410 W3-W0 seasonal ET-contrast RMSE was 45.79 mm for wheat and 66.84 mm for maize across three calibration
411 years; the absolute 2019 contrast errors were 116.37 and 52.74 mm. For grain contrasts of W1-W3 relative to same-
412 year W0, calibration RMSE was 0.519 and 1.669 t ha⁻¹ and retrospective-testing RMSE was 1.316 and 1.323 t ha⁻¹,
413 respectively, at 13% moisture (Table S5; Figure S2). Maize treatment labels denote preceding wheat irrigation;
414 maize irrigation was identical within each year.

415 **Table 3. Wuqiao water calibration and retrospective testing.**

Crop	Period	Target	n	Unit	RMSE	MBE	nRMSE (%)	NSE	R ²
Maize	2016–2018	Seasonal ET	12	mm	37.444	3.604	12.69	0.533	0.619
Maize	2016–2018	Grain	12	t ha ⁻¹	2.267	-1.738	18.78	-2.940	0.128
Maize	2016–2018	Biomass	12	t ha ⁻¹	4.688	-4.188	21.79	-5.625	0.073
Maize	2019	Seasonal ET	4	mm	55.637	-35.902	14.13	-0.346	0.343
Maize	2019	Grain	4	t ha ⁻¹	1.700	1.403	15.97	-3.641	0.972
Maize	2019	Biomass	4	t ha ⁻¹	2.038	0.726	10.44	-0.313	0.123
Wheat	2016–2018	Seasonal ET	12	mm	40.188	-12.649	8.97	0.689	0.758
Wheat	2016–2018	Grain	12	t ha ⁻¹	1.249	-0.445	16.74	0.746	0.817
Wheat	2016–2018	Biomass	12	t ha ⁻¹	4.222	-2.480	25.48	0.496	0.781
Wheat	2019	Seasonal ET	4	mm	73.805	48.929	21.98	0.503	0.915
Wheat	2019	Grain	4	t ha ⁻¹	3.161	3.108	83.27	-4.654	0.870
Wheat	2019	Biomass	4	t ha ⁻¹	5.427	4.908	55.66	-0.752	0.924

416 Calibration uses 12 treatment seasons per crop in 2016–2018; retrospective testing uses four in 2019. Counts are
417 treatment means rather than independent replicate seasons. ET is a 0–2 m water-balance estimate under negligible
418 runoff and drainage assumptions. Grain is at 13% moisture and aboveground biomass is dry matter. Annual grain
419 means and SE are digitized from source Supplementary Figure S2. Initial total storage is observed; layer
420 distribution and transferred hydraulics remain conditional.

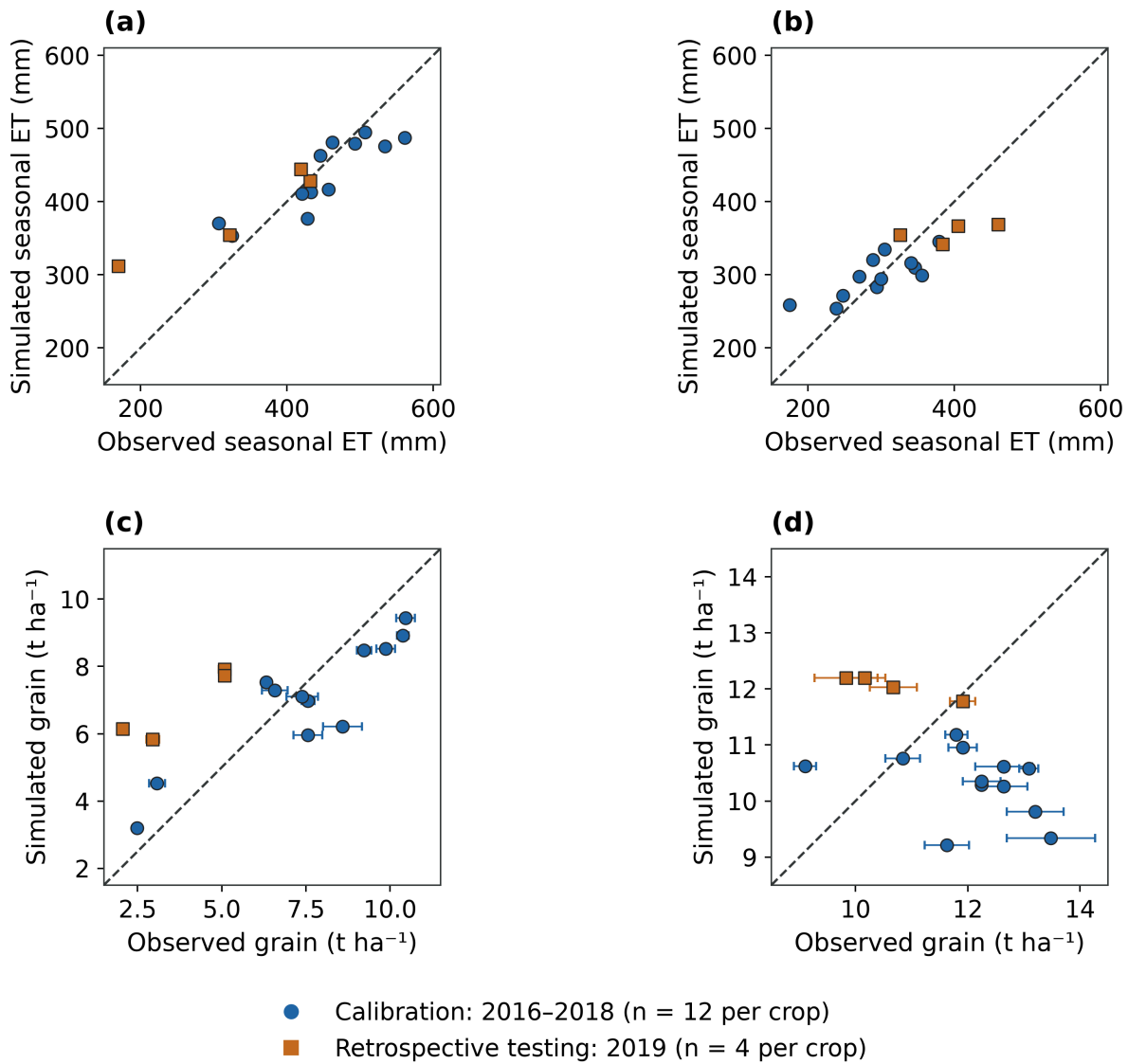


Figure 6. Wuqiao calibration and retrospective testing. (a,b) Seasonal ET; (c,d) annual treatment-mean grain at 13% moisture. Wheat occupies the left column and maize the right. Circles show 2016–2018 treatments used to fit water-use parameters; squares show 2019, withheld from the fit and selection. Dashed lines denote 1:1 agreement. Crop-specific grain axes focus on their observed ranges. Horizontal grain error bars are the published SE from three field replicates. Observed ET follows the published 0–2 m water balance; simulated dry grain is converted once by division by 0.87. Maize labels retain preceding wheat management with identical maize irrigation within each year.

3.3. Grain production and water balance under irrigation allocation

At a 50% regional irrigation reduction, targeted allocation gave a simulated annual rotation dry-grain yield of 15.761 t ha⁻¹ yr⁻¹ for wheat and maize combined, compared with 15.029 under uniform cuts. Both yields remained below the conventional yield of 16.756 t ha⁻¹ yr⁻¹. Targeting therefore retained 0.732 t ha⁻¹ yr⁻¹ more grain than

uniform allocation (4.87% of uniform production), equivalent to 7.219 million t yr⁻¹ across the fixed 9.867 million rotation hectares (Table 4; Figure 7). Targeted production remained 9.813 million t yr⁻¹ below conventional production.

Both allocations supplied 190 mm yr⁻¹ of field irrigation. Targeted and uniform crop-plus-fallow ET averaged 827.70 and 806.94 mm yr⁻¹, respectively. The grain advantage was accompanied by 20.76 mm yr⁻¹ (2.048 km³ yr⁻¹) greater ET. Profile drainage was 79.04 versus 96.34 mm yr⁻¹, and runoff was 14.43 versus 18.99 mm yr⁻¹ (Table 4). The targeted-minus-uniform grain contrast at the 50% cut was positive in all twelve comparison years (Figure 7c). The positive regional contrast was accompanied by spatially contrasting yield responses (Section 3.4).

Table 4. Simulated annual rotation production and water balance under equal irrigation reductions, 2014–2025.

Strategy	Dry grain (t ha ⁻¹ yr ⁻¹)	Irrigation (mm yr ⁻¹)	ET (mm yr ⁻¹)	Drainage (mm yr ⁻¹)	Runoff (mm yr ⁻¹)
Conventional	16.756	380.00	887.43	185.26	39.32
Uniform, 25% cut	16.177	285.00	856.41	135.94	25.71
Targeted, 25% cut	16.562	285.00	870.81	125.90	19.86
Uniform, 50% cut	15.029	190.00	806.94	96.34	18.99
Targeted, 50% cut	15.761	190.00	827.70	79.04	14.43
Uniform, 75% cut	13.347	95.00	741.57	68.72	15.83
Targeted, 75% cut	14.018	95.00	752.30	60.01	13.68

All per-hectare means use the same 9.866602 million mapped rotation hectares. Dry grain combines wheat and maize harvests; ET includes crops and fallows. Regional grain in million tonnes equals rotation yield multiplied by mapped area in million hectares. Regional water in km³ equals depth in mm multiplied by hectares and 10, divided by 10⁹. The conventional schedule supplies 300 mm to wheat and 80 mm to maize.

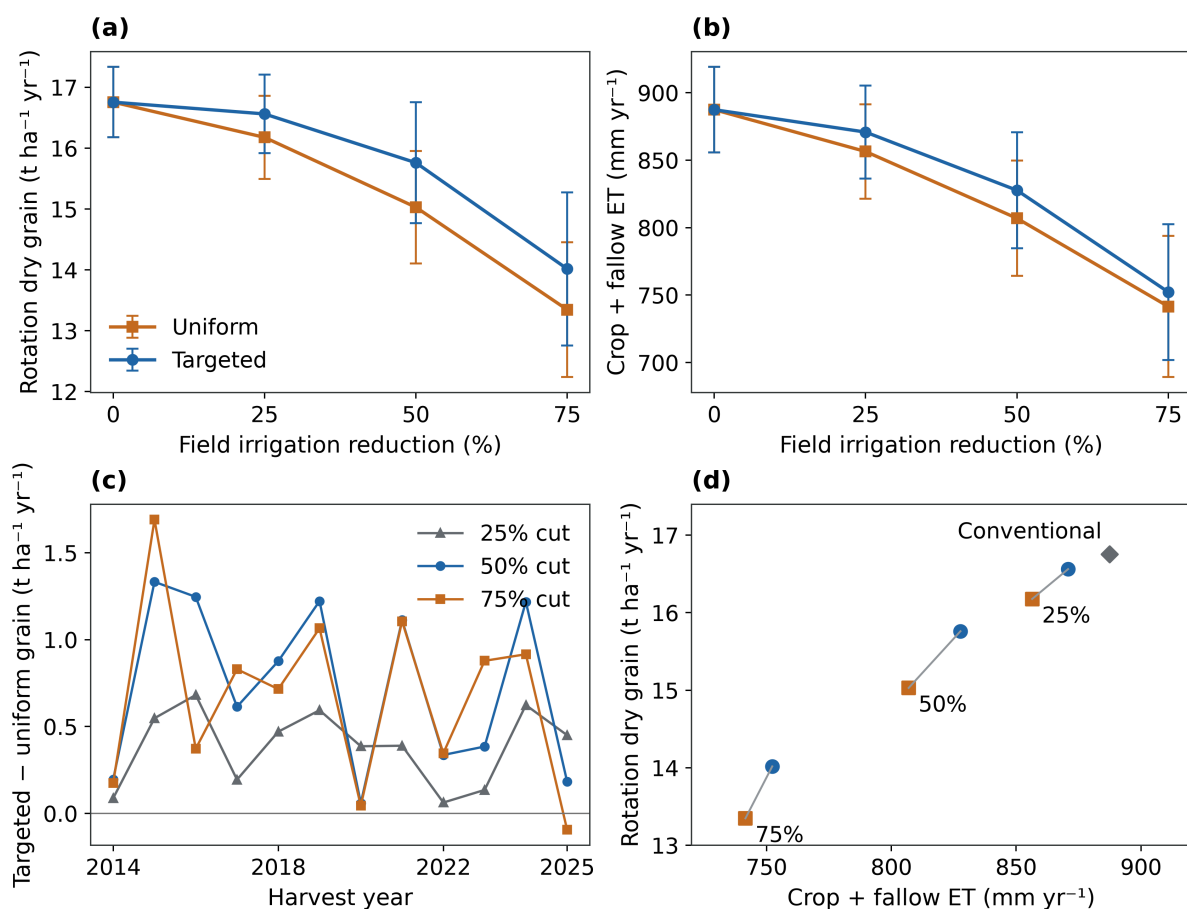


Figure 7. Irrigation allocation and grain–water trade-offs during 2014–2025. (a,b) Rotation-area-weighted annual combined wheat–maize dry grain and crop-plus-fallow ET; error bars are between-year SD. (c) Annual targeted-minus-uniform grain at identical irrigation budgets. (d) Period means; connectors join policies with the same budget. Values per hectare use the same 9.867 million mapped rotation hectares, rather than separate crop areas. Targeting retains more grain than uniform cuts while remaining below conventional production.

3.4. Spatial distribution and persistence of yield contrasts

Relative to uniform allocation, the central and northern latitude bands contributed grain gains of 7.475 and 1.886 million t yr^{-1} , while the southern band contributed a loss of 2.142 million t yr^{-1} . The southern band occupied 60.7% of mapped rotation area. Mean rotation-yield contrasts were +2.415, +2.420 and $-0.357 \text{ t ha}^{-1} \text{yr}^{-1}$ in the central, northern and southern bands, respectively (Figure 8; Table S9).

At the 50% regional irrigation reduction, targeted allocation increased mean annual combined wheat–maize dry grain relative to uniform reduction on 43.92% of the mapped rotation area (4.333 Mha), decreased it on 37.60% (3.710 Mha), and left it unchanged on 18.48% (1.823 Mha). Mean contrasts within the gain and loss areas were

457 +2.439 and -0.903 t ha⁻¹ yr⁻¹, respectively. Their regional contributions were +10.568 and -3.349 Mt yr⁻¹, giving a
458 net regional grain difference of 7.219 Mt yr⁻¹ (0.732 t ha⁻¹ yr⁻¹ across the full mapped area). Local losses offset
459 31.69% of the gross positive contribution (Figure 8d).

460 The area-weighted median grain contrast was 0.000 t ha⁻¹ yr⁻¹, and the 10th and 90th percentiles were -1.182 and
461 +3.553 t ha⁻¹ yr⁻¹. Relative to uniform reduction, mean annual field irrigation increased by 119.0 mm yr⁻¹ in the
462 gain area and decreased by 139.0 mm yr⁻¹ in the loss area. The corresponding crop-plus-fallow ET contrasts were
463 +94.4 and -55.1 mm yr⁻¹. Irrigation changes balanced at regional scale, whereas the area-weighted mean ET
464 contrast was +20.8 mm yr⁻¹.

465 Negative yield contrasts persisted over 2014–2025: 21.34% of the mapped rotation area had lower yields than under
466 uniform allocation in all 12 years, and 37.60% experienced a loss in at least one year. Every response unit with a
467 negative period-mean contrast recorded losses in at least 9 years; all response units with a positive period-mean
468 contrast recorded gains in all 12 years. Across years, the mean annual area exposed to a loss was 35.37%. Mapped
469 cells with negative period-mean contrasts formed 8 connected components under rook adjacency. The largest
470 contained 3.700 Mha of mapped rotation area (99.73% of all loss area), with occupied cell centers extending from
471 32.0 to 36.3°N and 112.8 to 121.5°E.

472 Negative period-mean yield contrasts occupied 32.80%, 37.60%, and 60.08% of mapped rotation area at reductions
473 of 25%, 50%, and 75%, respectively. At the 75% reduction, the area-weighted median contrast became negative
474 (-0.788 t ha⁻¹ yr⁻¹), although gains on 37.04% of the area produced a positive regional balance. Gross positive and
475 negative contributions at that reduction were +14.100 and -7.487 Mt yr⁻¹, leaving a net gain of 6.614 Mt yr⁻¹, below
476 the net gain at the 50% reduction.

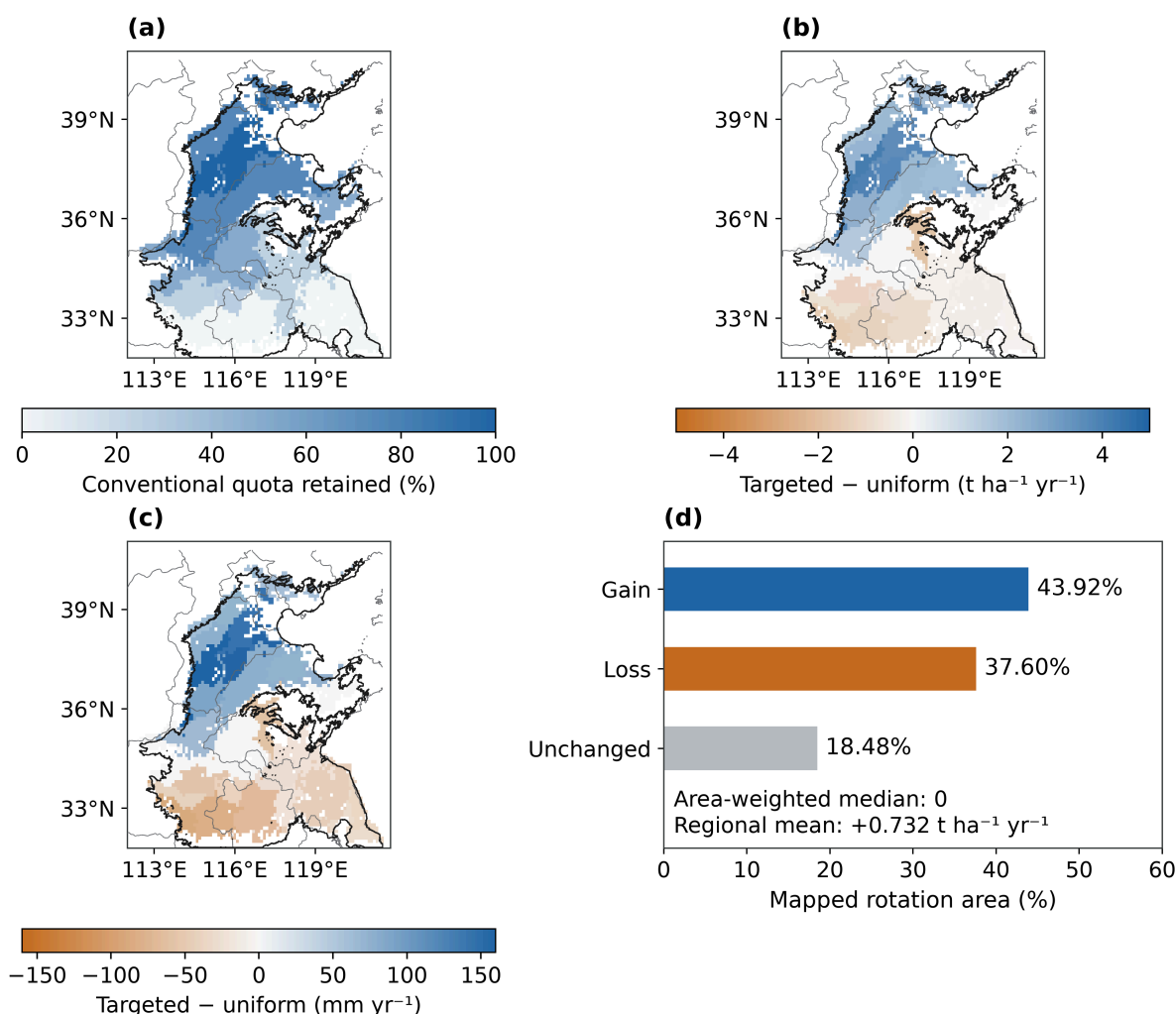


Figure 8. Spatial redistribution at a common 50% regional irrigation reduction, 2014–2025. (a) Targeted fraction of the conventional quota; uniform allocation retains 50% everywhere. (b,c) Targeted-minus-uniform combined rotation dry grain and crop-plus-fallow ET. (d) Mapped rotation-area shares with positive, negative or unchanged mean grain contrasts. The 3,641 cells inherit 32 representative responses. Thin grey provincial lines and black study-domain outlines follow Figure 2; unmapped areas remain blank.

3.5. Antecedent water availability and annual irrigation strategies

Four antecedent storage–rainfall classes occurred during 2003–2013. Nine comparison years had eligible GRACE observations, all in the lower-storage category; three years without eligible GRACE observations received conventional irrigation quotas. At the 95% grain target, both storage strata selected an irrigation fraction of 0.75 under lower rainfall and 0.50 under higher rainfall. The 98% target selected 1.00 and 0.75, respectively (Table 5). Continuous strategies selected at the 95% and 98% targets retained 96.27% and 99.63% of conventional grain production over the nine years with eligible GRACE observations. Field-irrigation reductions were 130.72 and 35.72 mm per rotation; corresponding ET reductions were 36.84 and 6.17 mm (Figure 9; Table S7). The 95% strategy retained at least 95% in 8 of nine years, with a minimum of 92.12%.

Storage–rainfall and matched rainfall-only strategies selected identical irrigation fractions and produced identical grain, ET, drainage and irrigation histories at both targets. Rainfall-only strategies that applied decisions during GRACE gaps followed different irrigation histories and carried different soil-water states into subsequent crops (Table S8).

Table 5. Selected irrigation candidates by relative water-availability class.

Class	Irrigation fraction	Wheat irrigation (mm)	Maize irrigation (mm)	Selection grain retention (%)
Higher storage dry	0.750	225.000	60.000	95.537
Higher storage wet	0.500	150.000	40.000	97.093
Lower storage dry	0.750	225.000	60.000	96.617
Lower storage wet	0.500	150.000	40.000	97.431

Fractions meet the 95% class-mean grain target during selection years 2003–2013. Current shared crop parameters apply throughout. Both storage strata select identical fractions within each rainfall group; higher-storage classes have no later-period exposure.

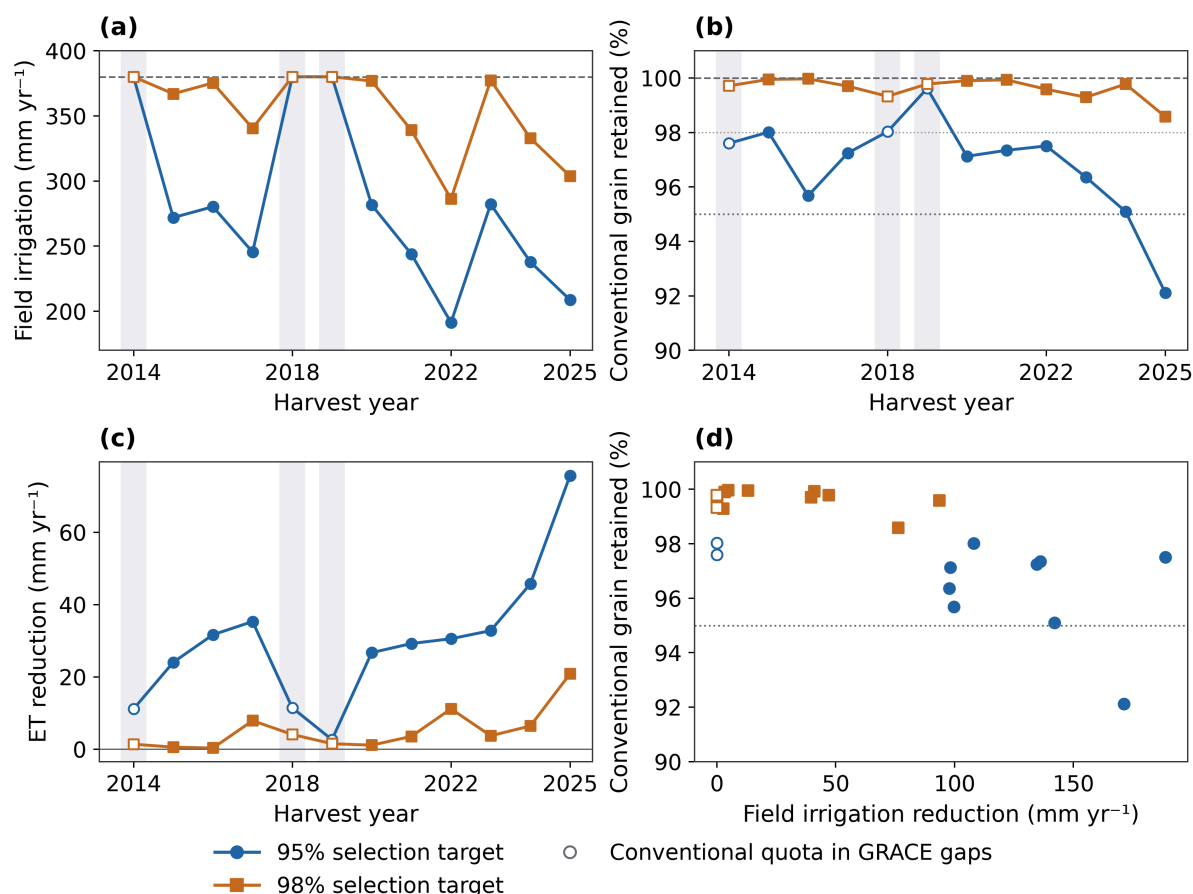


Figure 9. Continuous annual irrigation strategies during 2014–2025. (a) Field irrigation; (b) grain retained relative to conventional irrigation; (c) crop-plus-fallow ET reduction; (d) annual production–irrigation combinations. Blue circles and orange squares identify 95% and 98% selection targets. Shading and open markers identify missing-GRACE years with conventional quotas. Grain and ET can still differ in those years because preceding strategy decisions alter carried soil-water states. Matched rainfall-only histories coincide with the storage–rainfall histories.

4. Discussion

We used a calibrated crop model to simulate continuous winter wheat–summer maize rotations across the North China Plain, compare uniform and spatially targeted irrigation under equal water budgets, and test whether antecedent regional water storage improved decisions based on rainfall. Sustaining grain production under restricted irrigation requires consideration of both consumptive water use and the distribution of production losses across locations. The conditional results show that higher regional grain retention can accompany greater evapotranspiration and lower production in some locations, whereas storage information did not change the selected irrigation quotas within the tested design. The management value of these strategies therefore depends on accounting for the complete rotation water balance, soil-water carry-over across crops and fallows, and reliable crop responses to irrigation.

513 **4.1. Marginal irrigation returns and spatial allocation**

514 Spatial targeting exploits differences in marginal grain response to irrigation. Diminishing production gains across
515 field schedules (Yang et al., 2022) and spatial variation in irrigation requirements (Wu et al., 2019) provide a basis
516 for redistributing a limited supply, consistent with regional allocation studies (Jiang et al., 2016). The equal-budget
517 comparison attributes the simulated production difference to delivery allocation. Complete continuous histories
518 retain the effects of earlier irrigation and rainfall on subsequent crop responses.

519 At the 50% delivery reduction, targeting retained $0.732 \text{ t ha}^{-1} \text{ yr}^{-1}$ more combined wheat–maize dry grain than
520 uniform cuts. Across 9.8666 million rotation hectares, the corresponding reduction in grain loss was 7.219 million t
521 yr^{-1} , or 4.87% of the uniform total. Targeted production remained below conventional production. The positive
522 contrast in all twelve years indicates consistent regional gains within the simulated record. Its magnitude remains
523 dependent on the crop-response functions and declared management schedules (Table 4; Figure 7).

524 The spatial distribution sharpens this distinction between regional and local outcomes. Relative to uniform
525 allocation, mean losses occupy 37.60% of mapped rotation area, and losses persist in every comparison year on
526 21.34%. Gross gains of $10.568 \text{ Mt yr}^{-1}$ are partly offset by losses of 3.349 Mt yr^{-1} ; the area-weighted median is
527 zero despite a positive regional mean. The regional grain objective balances gains and losses among locations.
528 Local production floors constitute an additional allocation constraint.

529 Regional gain has a pronounced distributional cost. Central and northern band-mean rotation yields increased by
530 2.415 and $2.420 \text{ t ha}^{-1} \text{ yr}^{-1}$, while the southern mean decreased by $0.357 \text{ t ha}^{-1} \text{ yr}^{-1}$. The southern band covers
531 60.7% of the rotation area, so its losses substantially offset gains elsewhere (Figure 8; Table S9). The production
532 objective maximizes the area-weighted regional total and supplies no minimum yield for an individual location.
533 Local production floors and constraints on the distribution of irrigation reductions are therefore separate
534 requirements for allocation that protects affected farms.

535 **4.2. Rotation carry-over and the timing of water supply**

536 The spatial production response integrates water effects across both crops. Wheat irrigation can support canopy
537 growth and grain formation, replenish water available to maize or increase drainage. Reduced wheat delivery can
538 increase subsequent dependence on summer rainfall, producing different annual grain responses under different
539 rainfall sequences. The sixteen-year irrigation experiment shows diminishing production gains, and the Wuqiao

experiment links precipitation conditions with wheat scheduling and subsequent maize production (Yang et al., 2024; Yang et al., 2022). These coupled responses make the complete rotation the relevant unit for irrigation allocation.

Continuous soil-water histories preserve the delayed benefits and costs of allocation and annual quota changes (Figure S3). Production can be maintained by drawing down profile storage, with later rainfall or irrigation determining its replenishment. A finite-period delivery reduction can therefore include a contribution from stored soil water. Crop-plus-fallow accounting also captures evaporation and drainage between harvest and sowing. Their inclusion connects apparent production retention with the water remaining for the next crop and the complete rotation water balance.

The field evidence constrains only part of this carry-over mechanism. Wuqiao simulations use reported total profile storage with an assumed layer distribution. Wheat crop-start storage follows the pre-sowing simulation, whereas maize is independently initialized from treatment-specific observations. Maize irrigation is identical among treatments within each year, and its treatment labels identify preceding wheat schedules (Yang et al., 2024). Maize comparisons thus evaluate crop responses to the initial state associated with wheat management (Figure 6; Figure S2). Regional carry-over additionally depends on the model's continuous propagation and vertical redistribution of water between crops.

4.3. Production, consumptive use and groundwater accounting

The production benefit of targeting carries a consumptive-use trade-off. At the common 50% delivery reduction, targeting increased ET by $2.048 \text{ km}^3 \text{ yr}^{-1}$ and reduced profile drainage by $1.707 \text{ km}^3 \text{ yr}^{-1}$ relative to uniform allocation. With rainfall and total irrigation held constant, differences in ET are balanced by drainage, runoff and soil-storage change (Table 4). A shared delivery constraint therefore permits different consumptive outcomes. Irrigation productivity and consumptive-water savings require distinct objectives (Grafton et al., 2018; Perry et al., 2009).

The soil-profile balance also has a different boundary from the aquifer balance. Drainage below 2 m enters a deeper vadose zone, with storage and transit outside the modeled profile (Wu et al., 2023). Reduced field delivery or profile drainage therefore provides insufficient information to quantify changes in pumping, aquifer recharge or groundwater storage. Production and ET comparisons can identify agronomic trade-offs, while groundwater outcomes require the corresponding subsurface and abstraction accounting.

568 GRACE offers regional storage context at a broader scale. Its anomalies integrate groundwater with other terrestrial
569 stores, and its effective observation scale exceeds the displayed grid resolution (Save et al., 2016; Scanlon et al.,
570 2016). Recent aquifer recovery reflects water transfers, abstraction management and hydroclimatic conditions
571 (Long et al., 2025; Xu et al., 2026). The crop-model scenarios quantify changes in field delivery, consumption and
572 profile outflow without attributing those observed storage changes to individual irrigation strategies.

573 **4.4. When storage information changes an irrigation decision**

574 A storage indicator adds decision value when its inclusion changes the selected action or improves the grain–water
575 outcome under common constraints. GRACE represents regional storage conditions, while antecedent rainfall
576 describes more local water input (Save et al., 2016; Scanlon et al., 2016). Their different physical support motivates
577 joint classification, but decision value also depends on how the resulting classes distinguish among the available
578 irrigation fractions.

579 Both storage strata selected the same fraction for a given rainfall group at the 95% and 98% grain targets. Storage–
580 rainfall and matched rainfall-only strategies consequently produced identical complete histories of irrigation, crop
581 states, grain, ET and drainage. The storage split supplied additional environmental description without changing an
582 irrigation action in this five-level quota design. All nine later years with eligible observations fell in the lower-
583 storage category, so the higher-storage choices also lacked later-period exposure (Table 5; Figure 9). These features
584 delimit the configuration and conditions represented by the equal outcomes.

585 Monitoring coverage must remain distinct from the grouping criterion. Rainfall-available controls can change
586 irrigation during GRACE gaps, and those earlier decisions alter the soil-water state inherited in later years. Their
587 outcomes therefore combine rainfall-based selection with additional years of management. Matched coverage
588 preserves the appropriate comparison for the incremental effect of storage information. Finer quota choices,
589 alternative thresholds and verified release availability would define different decision experiments with their own
590 selection and evaluation requirements.

591 **4.5. Crop-response constraints and management reliability**

592 Management selection depends on the modeled marginal response to water, in addition to absolute grain and canopy
593 agreement. Intercepted-PAR radiation-use efficiency (Kiniry et al., 1989), cohort-specific leaf area and thermal
594 senescence connect canopy development with carbon supply; grain number and fixed individual grain mass
595 constrain the sink. Root uptake and the assimilation stress exponent then determine how water shortage changes

596 production. Variation among wheat-model transpiration formulations demonstrates the sensitivity of water-use
597 predictions to these processes (Cammarano et al., 2016). Water coefficients near their prescribed bounds remain
598 effective empirical estimates whose physiological interpretation and uniqueness are limited.

599 The estimation and evaluation data provide different evidence for these responses. Earlier growth estimates use 480
600 wheat and 433 maize observations across four wheat sites and five maize sites, with whole site-years and balanced
601 site contributions. Incomplete station irrigation histories limit transfer comparisons, and station observations have
602 no weight in the fitted water-response objective. Wuqiao 2016–2018 supplies the documented treatment seasons for
603 that fit. Wuqiao 2019 supplies a single test year with reported total storage, assumed layer distributions and
604 transferred soil hydraulics; prior inspection of its outcomes makes the test retrospective (Figures 4–6). Regional
605 comparison years also overlap crop-fitting years, so regional production differences characterize model scenarios
606 rather than time-forward crop validation.

607 The absolute regional production reference also depends on the prescribed management and crop parameters.
608 Conventional simulations produce 8.065 t ha^{-1} of wheat dry grain and 8.691 t ha^{-1} of maize dry grain per year.
609 These values represent fixed irrigation dates, crop calendars and nutrition modifiers rather than an independently
610 validated regional harvest inventory. The $0.732 \text{ t ha}^{-1} \text{ yr}^{-1}$ allocation advantage depends on the same simulated
611 marginal responses; its regional scaling does not remove field-response uncertainty.

612 Substantial field-test errors constrain confidence in irrigation-response transfer. Wheat grain nRMSE was 83.27%,
613 with a bias of $+3.108 \text{ t ha}^{-1}$ at the published 13% moisture basis; the absolute W3–W0 ET-contrast error was 116.37
614 mm. Maize ET NSE was negative, with an ET-contrast error of 52.74 mm (Table 3; Table S5). These discrepancies
615 affect the response functions used to select regional allocations. Numerical water-balance closure demonstrates
616 conservation but supplies no measure of response accuracy. The eight purposive actual-cell checks address a
617 separate uncertainty, spatial aggregation, and provide no regional field-validation sample. Depth-resolved soil water
618 and complete dated management would better constrain uptake and redistribution; canopy and soil-moisture
619 assimilation also requires treatment of retrieval errors and observation operators (Ines et al., 2013).

620 Production-retention targets introduce a further management distinction between period means and annual
621 reliability. Mean retention of 96.27% and 99.63% coincided with annual minima of 92.12% and 98.58%; the 95%
622 strategy met that threshold in eight of nine class-available years (Figure 9). A mean target consequently supplies no
623 shortfall probability or guarantee for an individual farm. Local production floors, acceptable shortfall frequency and

624 a reserve entering the next crop can complement regional delivery and consumption constraints. Forecast-based
625 irrigation studies illustrate explicit treatment of water-stress risk (Roy et al., 2021). Field scheduling additionally
626 requires current soil-water information, weather forecasts and application timing.

627 **4.6. Limitations and future directions**

628 The principal limitation is uncertainty in the crop responses used for strategy selection. Field testing covers a single,
629 previously inspected year, while incomplete station management and assumed initial water profiles constrain
630 transfer. Remaining errors in ET and grain responses prevent the simulated policy advantages from being
631 interpreted as independently validated field benefits. Further evaluation requires new site-years with complete
632 management records, withheld from both model development and strategy selection. Concurrent measurements of
633 soil water by depth, ET, canopy development and harvest would allow separate assessment of water redistribution,
634 crop stress and irrigation-treatment contrasts.

635 Regional results also depend on a static rotation map, representative climate–soil units, prescribed irrigation dates
636 and fixed nutrition modifiers. Checks at purposively selected source cells cannot bound regional uncertainty.
637 Dynamic nitrogen cycling is absent, and changes in crop area or actual farm practices are not reconstructed. Future
638 analyses should expand evaluation at individual source cells and propagate uncertainty in crop parameters, soil
639 hydraulics, weather and spatial aggregation through paired policy histories. Allocation stability and the magnitude
640 and persistence of local losses require uncertainty assessment alongside the regional mean.

641 The storage comparison is specific to the tested quota grid, classification thresholds and lower-storage conditions
642 represented during later comparison years. GRACE release availability before sowing remains unverified, so the
643 scenarios do not establish operational scheduling performance. Prospective decision experiments should use
644 information available on each decision date, maintain matched rainfall-only controls and evaluate both storage
645 states. Finer quotas and application timing informed by soil water could be tested with constraints on consumptive
646 use, local production and the risk of production shortfalls.

647 Finally, the profile uses free drainage at its lower boundary, with no capillary supply or deeper subsurface storage,
648 and field irrigation is not linked to measured abstraction. Groundwater applications require joint accounting of soil
649 water and aquifer flows, constrained by observations of pumping, recharge and storage change. Delivery capacity,
650 irrigation costs and farmer participation also require assessment before the agronomic scenarios can support
651 operational management.

652 **5. Conclusions**

653 At an identical 50% irrigation reduction, targeted allocation retained 0.732 t ha⁻¹ yr⁻¹ more simulated combined
654 wheat–maize dry grain than uniform cuts during 2014–2025. This reduced regional grain loss by 7.219 million t
655 yr⁻¹ (4.87% of uniform production), while increasing crop-plus-fallow ET by 20.76 mm yr⁻¹ (2.048 km³ yr⁻¹).
656 Relative to uniform allocation, mean grain losses occurred on 37.60% of mapped rotation area and persisted in all
657 twelve years on 21.34%. Regional production retention therefore requires separate consumption and local
658 production constraints.

659 Annual strategies selected at 95% and 98% grain-retention targets retained 96.27% and 99.63% of conventional
660 production over nine years with eligible GRACE observations. The 95% strategy reduced irrigation by 130.72 mm
661 and ET by 36.84 mm per rotation. Matched rainfall-only controls selected identical quotas and produced identical
662 outcomes, so storage classification provided no additional decision benefit within the tested quota design and
663 monitoring coverage.

664 Continuous rotation accounting preserves the influence of preceding management on subsequent production and
665 water use. The field-test errors and incomplete station management delimit confidence in transferring the estimated
666 crop responses. Translation of these regional scenarios into irrigation management depends on observed response
667 accuracy, explicit production-risk constraints and field scheduling informed by soil water and forecasts.
668 Groundwater effects additionally require pumping, recharge and deeper subsurface storage beyond the crop-model
669 boundary.

670 **Data and code availability**

671 Model code, frozen calibration and regional inputs, observation subsets and quality exclusions, weather, calibrated
672 parameters, predictions, evaluation metrics and figure-data tables are available at [https://github.com/SmartAG-](https://github.com/SmartAG-Team/agri_water_management)
673 [Team/agri_water_management](https://github.com/SmartAG-Team/agri_water_management). The repository includes a standalone Open Crop Model package, pinned
674 dependencies, reproduction commands and SHA-256 provenance. Reproduction uses the included inputs without
675 another model checkout or access to the historical workspace. Original datasets retain their provider licences and
676 attribution requirements, documented in the repository source catalogue.

677 **Declaration of generative AI and AI-assisted technologies**

678 AI assisted with grammar checking.

679 **References**

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